Lab 5: ARMA estimation, diagnostics and forecasting

Box–Jenkins on simulated and real series

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Goals

- Identify candidate ARMA orders from sample ACF/PACF.
- Estimate AR, MA and ARMA models in `statsmodels`.
- Compare candidates by AIC and BIC.
- Validate the chosen model with Ljung–Box on residuals.
- Produce point forecasts and confidence intervals.

Setup

- Python 3 with `numpy`, `matplotlib`, `statsmodels`.
- `numpy.random.seed(42)`.
- Data: `data/AvTempAtlanta.txt` (Chapter 2/3 dataset).

Exercise 1 – AR(2) simulation and estimation

(1) Simulate $T = 500$ observations of

$$X_t = 0.6\, X_{t-1} - 0.3\, X_{t-2} + \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

Plot the series, the sample ACF, and the sample PACF up to lag 20.
(2) Read the PACF: at which lag does it cut off? What order does this suggest?
(3) Fit `ARIMA(order=(2, 0, 0))`. Report $\widehat{\phi}_1$, $\widehat{\phi}_2$, their standard errors, and the AIC. Compare with the true values $(0.6, -0.3)$.
(4) Repeat with order $(1, 0, 0)$ and $(3, 0, 0)$. Use AIC to confirm that $(2, 0, 0)$ is best.

Exercise 2 – MA(1) identification

(1) Simulate $T = 500$ of

$$X_t = \varepsilon_t + 0.7\, \varepsilon_{t-1}, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

(2) Plot ACF/PACF up to lag 10. Verify that $\widehat{\rho}(1) \approx \theta/(1+\theta^2) = 0.7/1.49 \approx 0.470$ and $\widehat{\rho}(h) \approx 0$ for $h \geq 2$.
(3) Fit `ARIMA(order=(0, 0, 1))`. Report $\widehat{\theta}$ and AIC.
(4) Fit `ARIMA(order=(1, 0, 0))` on the same data. Why does the AR(1) fit fit slightly less well? Explain in two sentences using the invertibility argument.

Exercise 3 – ARMA$(1, 1)$ on simulated data

(1) Simulate $T = 500$ of
$$X_t = 0.5\, X_{t-1} + \varepsilon_t + 0.4\, \varepsilon_{t-1}.$$

(2) Fit four candidates: $(1, 0, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(2, 0, 2)$. Build a comparison table with columns: order, log-likelihood, AIC, BIC, Ljung–Box(24) p-value.
(3) Which candidate wins on AIC? On BIC? Why does $(2, 0, 2)$ over-fit?

Exercise 4 – Box–Jenkins on the Atlanta residual

Reload the Atlanta residual obtained in Lab 3 (centred MA $k = 6$ + seasonal averaging on `data/AvTempAtlanta.txt`).
(1) Plot ACF/PACF of the residual up to lag 36.
(2) Fit AR(1), AR(2), MA(1), ARMA$(1, 1)$ and ARMA$(2, 1)$. Report AIC, BIC, σ^2, Ljung–Box(24).
(3) Pick the best model on the (AIC, BIC, Ljung–Box) triplet. Justify in 5 lines.

Exercise 5 – Forecasting the chosen model

(1) On the chosen model from Exercise 4, produce a 24-step forecast. Plot the in-sample fit and the forecast with a 95% confidence band.
(2) How does the CI width evolve with the horizon? What limit does it converge to? Express that limit in terms of $\widehat{\sigma}$ and the ψ_j coefficients.
(3) [**bonus**] Fit the same model on the first 80% of the data and forecast the last 20%. Compute the RMSE on the hold-out. Comment.

Warning

The function `ARIMA(order=(p, 0, q)).fit()` returns an object whose `.summary()` prints the table you need; `.params` stores the coefficient vector; `.aic`, `.bic`, `.llf` hold the criteria; `.resid` stores the in-sample residuals.